Designing Sexual Communication Tools for and with Transgender and Queer Youth

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Sexual communication is critical to having healthy relationships and addressing sexual health inequities, but there are barriers to openly sharing clear information with one's partner(s), especially for transgender and queer youth. Technology-based interventions offer strategies for mediating sexual communication; however, sexual communication literature is overly focused on contraceptive use, and little work has explored how to identify effective strategies through community-based participatory research. We therefore describe a multi-year design partnership with a youth advisory board made up of 18 United States-based transgender and queer youth that yielded two technology-based sexual communication tools. Desire Dialogues facilitates reflection and preparation to discuss one's sexual boundaries with potential partners, and the Intimate Disclosure Toolkit guides users who wish to disclose personal information with others. By documenting our community- and value-driven design process, we identify participatory challenges, five design principles, and an agenda for future research that can guide the next generation of technology-based sexual communication tools.

CCS Concepts: • **Human-centered computing → HCI theory, concepts and models**; • **Social and professional topics → Sexual orientation**.

Additional Key Words and Phrases: Trans health, Queer health, Community-based research, Sexual Communication

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1 Introduction

Sexual health among youth has been a focus of significant interventional effort in the United States (U.S.). Of the reported U.S. cases of Sexually Transmitted Infections (STIs) in 2023, 48.2%—nearly half—were among adolescents and young adults ages 15–24 [39]. Given the interpersonal nature of relationships, communication plays a critical role in how young people safely navigate sexual and romantic interactions. Previous work has shown that sexual health communication between partners is linked to increased and consistent condom and contraceptive use, demonstrating an opportunity to leverage sexual communication to address sexual disparities among youth [93, 114].

In an ideal scenario, a young person who tests positive for an STI shares this information with their sexual partner(s), and everyone who has been exposed receives treatment. However, sexual health communication is not so straightforward, as social and structural factors like stigma prevent young people from sharing critical information [69, 89]. That same youth from the previous example might choose to not communicate with their partners to avoid the shame and judgment of others, leaving their sexual partner(s) at risk of spreading an STI.

To address these challenges, scholars have begun to explore how online interventions can facilitate sexual communication between youth and their partners. There are numerous benefits to using technology-based interventions for sexual health promotion: youth already use technology as a way to discuss their sexual health with their partners [114], and technology can reach a more expansive audience, provide standardized information in a customized way, and preserve privacy and autonomy [19, 34, 41, 76, 102]. Social Computing scholars have built upon these advantages to design for safer sexual communication, including supporting the disclosure of sensitive information [83] and detecting potentially harmful online interactions [54, 88].

While sexual health is a universal experience, this paper focuses on the design process of sexual health communication tools for one particular group: transgender (trans) and queer youth. Trans and queer young people face significant sexual health disparities, including higher rates of STIs and intimate partner violence [9, 28, 44, 90, 99, 103, 113], due to unique stigmatic social factors that inhibit their capacity to safely navigate sexual and romantic relationships. If the young person from the aforementioned example were trans, they may have additional difficulty openly communicating with their partner(s) for fear of being outed or harmed. In this paper, we sought to build upon trans and queer youth-focused, sexual health technology-based intervention research (e.g., [62, 73, 80, 100]) to develop online interventions that support safe sexual health communication.

To understand and address trans and queer youths' sexual health communication needs, we engaged in a multi-year partnership with a youth advisory board (YAB) made up of 18 trans and queer U.S.-based youth. YABs' strengths lie in establishing trust and deep community relationships through long-term engagements, making them well-suited for health and safety research with youth [63] [2, 109]. Drawing from traditions in trans-centered, community-based research [29, 48, 50, 62], establishing design partnerships with the YAB was critical to shifting decision-making power towards those affected by design choices and creating interventions that meet community needs [16, 29, 51]. Grounded in this YAB partnership, we explored the following research questions:

> RQ1: What sexual health communication technologies do trans and queer youth envision in promoting sexual health behaviors?
> RQ2: What values can support designers of technology-based interventions for trans and queer youths' sexual health communication?

From reporting our reflections, this work makes the following contributions centered around our co-design processes:

(1) We describe our iterative, community-driven design process—including brainstorming, prototyping, and evaluating—of two technology-based interventions that seek to promote sexual communication among trans and queer youth: Desire Dialogues and the Intimate Disclosure Suite.
(2) Next, we identify five design principles that can support the development of future sexual health communication technology-based interventions for trans and queer youth: approaching safety through a non-prescriptive, informed consent model; translating a joyful user experience into joyful signals; reinforcing individual strengths and readiness; support for real-world possibilities; and suspending designers' assumptions.
(3) Finally, we discuss how embedding desire-based frameworks into collaborative approaches and scaffolding transformative reflection through interactive, joyful experiences can support sexual communication among trans and queer youth, thereby advancing efforts to improve sexual health through technology-based interventions.

2 Related Work

This work builds upon three bodies of research: sexual communication between youth and their partners, developing online interventions for sexual health, and designing trans and queer technologies through community-based research.

2.1 Sexual Communication Between Youth and Their Partners

Sexual communication encompasses both the behavior of exchanging verbal and non-verbal messages and more social-cognitive aspects of communication, such as self-efficacy and fear of sharing information [115]. Greene and Faulkner [42] identify four types of sexual communication among heterosexual couples: discussions of sexual limits, discourse on pleasure, conversations about health, and safer sex talk. While Social Computing work has not explored these particular aspects of sexual communication, there has been much work focused on how specific dating platform features facilitate self-disclosure experiences [36, 82–84, 108].

A growing body of work suggests that greater sexual communication is linked to more, consistent condom and contraceptive use—critical behaviors that can reduce sexual health inequities [114, 116]. Those who communicating about sexual topics with their partners additionally report greater satisfaction in their relationships [68, 71, 117]. Despite these benefits, however, young people struggle to talk openly with their partners. Not only do they lack good models for this behavior from both media and their parents [38, 55] but also interpersonal barriers such as power dynamics and low confidence to navigate embarrassment or awkwardness lead to less communication [11, 47]. A 2014 study found that 54% of youth surveyed did not discuss sexual topics with their dating partners [114]. These benefits and challenges emphasize the importance of supporting young people with discussing important sexual topics with their partners.

Building off this foundational literature, this work extends sexual communication between youth and their partners in several ways. First, we expand sexual communication beyond a sole focus on condom and contraceptive use and towards relational aspects of sexual health. Drawing from previous literature examining sexual health among trans and queer youth [52, 62, 77], this includes both young people's internal perceptions (e.g., body image, gender dysphoria) and social dynamics (e.g., consent, sexual desire). A scoping review of youth sexual communication literature found that no studies focused on communication of consent and sexual pleasure [115]. Further, scholars have characterized sexual communication among American Indian youth [93] and African American girls [94], but there have been no studies examining trans and queer youths' sexual communication needs. Finally, we explore how to support trans and queer youths' sexual communication with their partners through technology-based interventions. While research suggests that technology offers promising benefits to promoting healthy behaviors [116], this work presents the first exploration into how to actually design these kinds of systems. In the next section, we outline what technology-based interventions can offer to sexual communication and sexual health promotion more broadly as well as the challenges of doing so effectively.

2.2 Technology-Based Interventions for Sexual Health Promotion

Online interventions have emerged as a mechanism for supporting sexual health needs and promoting healthy behaviors [1, 61, 62, 76]. For instance, Tortolero et al. [102] describe an intervention meant to delay sexual behavior among youth that combined "group-based classroom activities with computer-based instruction and personal journaling". Kiene and Barta [56] deployed a computer-delivered intervention targeting the increase of HIV risk-reduction behaviors. In addition, Bauermeister et al. [10] developed "imi", a web-based application meant to improve mental health by supporting LGBTQIA2S identity affirmation, coping self-efficacy, and coping skill practice. Previous work has highlighted the benefits of pursuing technological interventions, including their ability to reach a more expansive audience, provide standardized information in a customized way, preserve privacy and autonomy, and facilitate engaging, interactive, and personalizable learning experiences [19, 34, 41, 61, 76, 102]. These advantages are especially valuable to trans and queer youth; online spaces have long been a refuge for them by providing space for identity exploration, community support, and normalizing stigmatized experiences [33, 62].

Despite these benefits, designing online health interventions that lead to effective, long-term outcomes is difficult to achieve for several reasons. First, efforts must balance addressing not only an individual's decision-making but also the broader social forces that inform their behaviors (e.g., stigma, discrimination) [78]. Scholars have indicated that interventions that ignore these social influences and solely focus on individual behavior changes are less effective for marginalized people [105]. Additionally, trans and queer communities have a painful history of participating in technology and medical research that has been exploitative, necessitating trust-building and practices that protect their well-being and data privacy and security [5, 48, 87]. Scholars have pointed out that when interventionists ignore such histories, they can create interventions that end up exacerbating the very inequities that they seek to mitigate—known as intervention-generated inequities [106].

Critical health and design perspectives have thus advocated for technology-based interventional approaches that go beyond just the individual level of behavior change and engage with communal and environmental factors, such as stigma, homophobia, and transphobia [12, 65]. Veinot et al. [105] offer a model of targets for health informatics interventions across individual, community, and structural levels. As they suggest, technology has the potential to inform the community and structural levels in many ways, including mediating communication and supporting decisions [105]. Recent work in HCI has pursued more upstream factors by understanding how technology shapes sexual health experiences. For instance, investigations have revealed how memes on Tiktok work to destigmatize sexual health taboos [110], social media activity can facilitate sex education and sexual health promotion [121], and multiplayer mobile games can promote sexual health [120]. A growing body of Social Computing work has sought to combat online sexual risk experiences at the platform level through detection algorithms [3, 88]. More recently, emerging, AI-based technologies have begun to change the landscape of technology-mediated sexual health support [31, 57, 79]. For example, Wester et al. [112] explored how social robots can support young people in communicating health information with their parents. However, while these new technologies may offer important benefits in the future, their current forms require improvements. A scoping review of sexual health conversational agents revealed that most chatbots lack safety features and fail to properly account for trans and queer people's experiences [79].

Drawing upon lessons across technology-based intervention literature, we extend these efforts by designing online tools that are grounded directly from trans and queer youths' inputs. Doing so provides another approach to designing for sexual safety that can work alongside AI-based systems. We outline how to design trans and queer technologies through value-based and community-driven research in the following section.

2.3 Designing Trans and Queer Technologies Through Community-Driven, Value-Based Research

HCI Scholar Oliver Haimson provides a two-part definition of trans technologies: "trans technologies are technologies built or adapted to address the specific concerns and needs of trans people and communities [...and] are technologies that embody or support themes or characteristics of transness [48]. Haimson and colleagues have further pinpointed qualities of transness—temporality, openness, change, separation, realness, intersectionality, and erotics—and queerness—multiplicity, fluidity, and ambiguity—that are "at the heart of transgender experiences" and therefore crucial to the design of trans technologies [49]. HCI research has thus pursued the design of trans technologies (e.g., [25, 26, 50, 62, 98]) through methods that emphasize the relational, liberatory ethos of transness and queerness, in community-driven and value-based research.

Community-driven research offers a method for involving the people affected by design decisions into the development process, and it has become a popular approach for designing trans and queer technologies [22, 27, 50]. One investigation brought queer men together to co-design social platform features, demonstrating ways to provide users with more agency [4]. Delmonaco and colleagues also designed an LGBTQ+ health resource website, drawing from participatory methods to ensure they were meeting community needs [29]. Hardy et al.'s study involving participatory design workshops with LGBTQ+ people living in rural Midwestern United States uncovered how people must navigate challenges of visibility, safety, and access to resources and seek out sociotechnical systems to support their goals [53]. As these studies show, participatory methods' emphasis on relationship-building provides deeper insight into people's and communities' needs, strongly aligning it with social justice-oriented goals.

In addition, trans and queer HCI scholarship has also drawn upon values to guide the design process, often exerted through the framework of Value-Sensitive Design (VSD) [3, 40]. A value-based perspective provides insights into how users make decisions and perceive platforms as well as how designers embed motivations into design choices [107]. In doing so, value-based sense-making provides a tool for aligning disparate priorities, a key factor in designing human-centered systems [14]. As VSD has evolved, scholars have called for more localized value discovery based on community lived experiences, as opposed to pinpointing a universal set of values—aligning VSD with participatory methods [15, 60]. Value elicitation exercises, which identify motivations and reasons that support design decisions, are additional approaches that can reveal alignments and misalignments between the perceived values of a platform and those of their users. Previous work has utilized value elicitation to identify potentially discordant values (e.g., [14, 18, 30, 58, 123]). For example, DeVito et al. [30] report how social platforms push content that their trans and queer users find unhelpful and even harmful. By identifying misalignment, the authors identified two design-based values of LGBTQ+ social platform users, self-determination and inclusion [30]. Identifying people's values, therefore, allows designers to be better positioned to address the challenges trans and queer users face.

This paper, then, draws upon each of these design-based traditions in community-driven participatory research, value-driven design approaches, and a trans technology ethos. In documenting our own design process and resulting value-based reflections, we exhibit a case study meant to provide further guidance for future work invested in advancing the health and well-being of trans and queer people.

3 Methods

3.1 Design Partners

This project is part of a multi-year long engagement with a youth advisory board consisting of trans and queer youth from across the United States. The YAB was made up of 18 members, whose ages ranged from 15-25 at the beginning of our design partnership. YAB members resided in 13 U.S. states—7 of which have laws or policies that ban gender-affirming care for people under 18 as of 2025 [20]. We recruited YAB members via a national recruitment strategy including snowball sampling, contacting trans and queer youth-serving nonprofits, and our own networks. YAB members were paid $50 for every monthly meeting they participated in. This study received IRB approval from our institution's Institutional Review Board.

To support the YAB, three teams collaborated to identify design needs and translate insights into prototypes. First, the lead author acted as the *research facilitator* by managing interactions among the team network. The relationships and trust built over the multi-year collaboration between the YAB and lead author were foundational to the project's success. They convened YAB meetings, mediated communication between the design and content development teams and the YAB, and synthesized YAB insights into design needs. Next, the *design team* was made up of a master's student and contracted UX designer with extensive experience converting design needs into prototypes. They translated YAB ideas and concepts into wireframes and high-fidelity prototypes and documented how decisions were grounded in YAB insights. After the YAB provided feedback on the prototypes, the design team made changes accordingly. Third, the *content development team* was made up of four sexuality and wellness educators with expertise in developing youth sex education training. They assembled the information and copy for the prototypes (e.g., survey questions, scenarios, scripts), drawing from the generated syntheses provided by the lead author.

3.2 Study Design

3.2.1 Participatory Design Activities. Between September 2023 through September 2025, the YAB participated in monthly meetings to co-design concepts, provide feedback, and guide research decisions such as recruitment strategies. Following the Asynchronous Remote Communities method [64, 122], most meetings were held synchronously over Zoom with an asynchronous participation option via Discord for those who could not join. We drew upon Google Slides to facilitate participation, document feedback, and share materials. Meeting activities were spread across three phases: needs assessment, prototype development, and evaluation.

The *needs assessment phase* drew inspiration from existing techniques for designing with youth (e.g., [45]) and consisted of four steps: familiarizing, brainstorming, sketching, and responding. To familiarize YAB members with the landscape of existing tools, they first sought out their favorite apps or websites that they had used before, including existing sexual health apps. As part of our goal to understand interactivity, we prompted them to consider the following questions: What makes them memorable? How do you interact with these platforms (e.g., typing things into the search bar, scrolling through a page, taking quizzes or answering questions about yourself)? We then moved on to brainstorming potential technology-based solutions. To support this process, we drew from previous literature that identified six categories of sexual health information that trans and queer young people seek: puberty, STI prevention, fertility & pregnancy, contraception, desire and pleasure, and consent [52]. Previous work has also identified four ways trans and queer youth prefer to learn about these areas: Q&A, written text, video, and in-person [62]. For each of

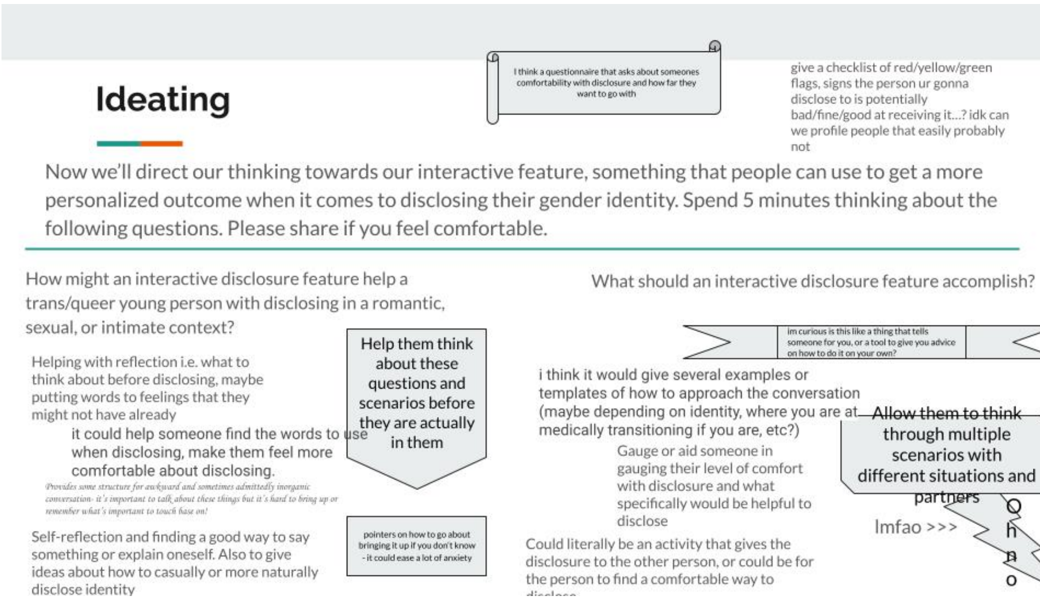


Fig. 1. Example of how YAB members brainstormed potential technology-based solutions to disclosing to their intimate partner(s).

the six categories, we asked YAB members to brainstorm feature ideas and what ways of learning they might draw upon. We encouraged them to think imaginatively and not limit themselves to existing technologies. We also sought to understand the purpose of potential technology-based solutions at this stage by discussing how specific ideas might help trans and queer youth and what they should accomplish, seen in Figure 1. Next, YAB members reviewed the list of feature ideas generated by the group YAB members, then chose three feature ideas to create rough sketches for what they might look like and how they might operate. Finally, YAB members then reviewed the group's sketches and responded to each with the following questions: 1) What do you like about this idea? 2) Is there anything you would add to this idea? 3) What do you think you could learn from interacting with this idea?

Transitioning from the open-ended nature of the previous phase, the *prototype development phase* sought to identify specific online tools to proceed with, including their purpose and functionality. The research team synthesized the ideas generated from the brainstorming and sketching steps into six potential tools, which we document in Section 3.3. After asking YAB members to rank them in order of preference, we chose to move forward with the top two concepts: a survey to facilitate communication of sexual preferences, boundaries, interests with partner(s) and a simulation for disclosure of identity. Once the YAB agreed to this direction, we sought to understand the context for how potential users might engage with each tool. They created storyboards to outline who is involved, how they use it on the platform, and what they do with it during or after. YAB members could create storyboards visually or with text on provided templates to allow for flexible participation options. From the storyboards, the design and content development teams created an initial sets of low-fidelity prototypes, including visual and interactive elements and sample copy.

Finally, in the *evaluation phase*, the lead author presented the design and content development teams' work to the YAB, then facilitate discussion around what they liked or disliked. We prompted feedback through questions such as: Does this align with your vision of this idea? Do the flow and

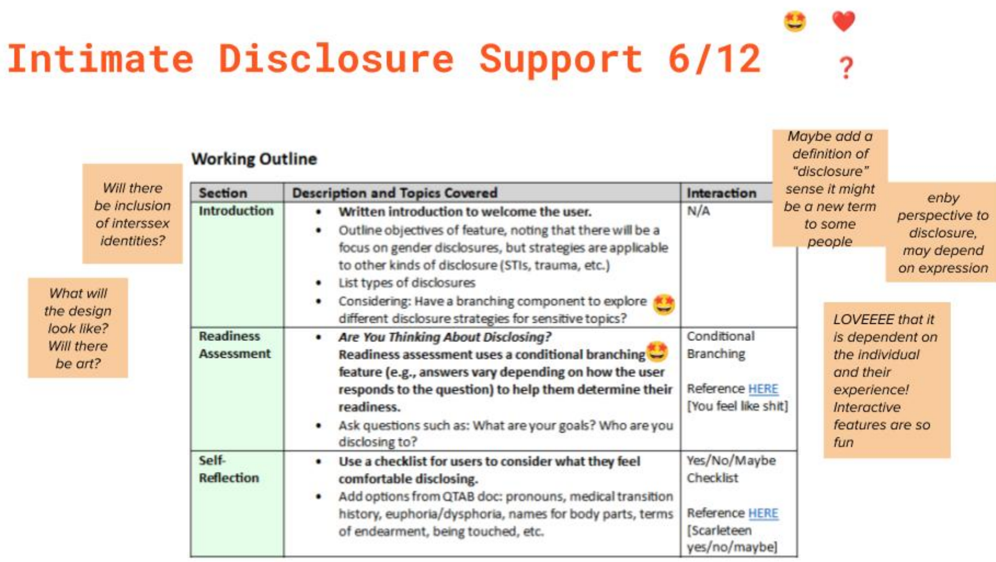


Fig. 2. Example of how YAB members provided feedback during meetings. The YAB provided feedback on a proposed feature based on previous ideas.

context make sense? What's missing? What is too much? Each monthly meeting provided space for the research team and YAB to weigh decisions that the other teams made. Figures 2 and 3 provide examples of how YAB members provided feedback on the Intimate Disclosure Toolkit described in Section 4.2. After soliciting feedback, the design and content development teams iterated upon the visual, interactive, and written elements.

3.2.2 Retrospective Value Elicitation. After the development process, we engaged in a retrospective value elicitation exercise with the content development and design teams. Scholars have previously drawn upon design-based artifacts to draw out value reflections, often focusing on patient or participant perceptions of established work [91, 118]. We similarly drew upon the established prototypes and asked the design and content development teams who developed them to engage in an exercise to identify values that were embedded into their decision-making. While value elicitation is typically conducted as part of end-user evaluation, we enacted a retrospective reflection with the content and design teams to identify design values for future development.

Through a semi-structured focus group with the design and content teams that lasted one hour each, we proceeded with the following steps. We first discussed the overall approach taken for each member in participating in this project. We sought out their initial understanding of the project and its goals, what they perceived to be unique about this project compared to others in the past, and where they drew inspiration from when developing content. Next, for the value elicitation exercise, we collaboratively worked on the virtual workspace FigJam. After first establishing a shared understanding of the goals of the exercise, each member of the team took ten minutes to review the content and design work they had created. As they reflected, we asked them to identify values that informed how they made decisions. For each value, we then asked them to specify value enactments or how each value manifested in either a procedural, design, or content decision (or non-decisions). Finally, the team members reviewed each other's values and respective enactments

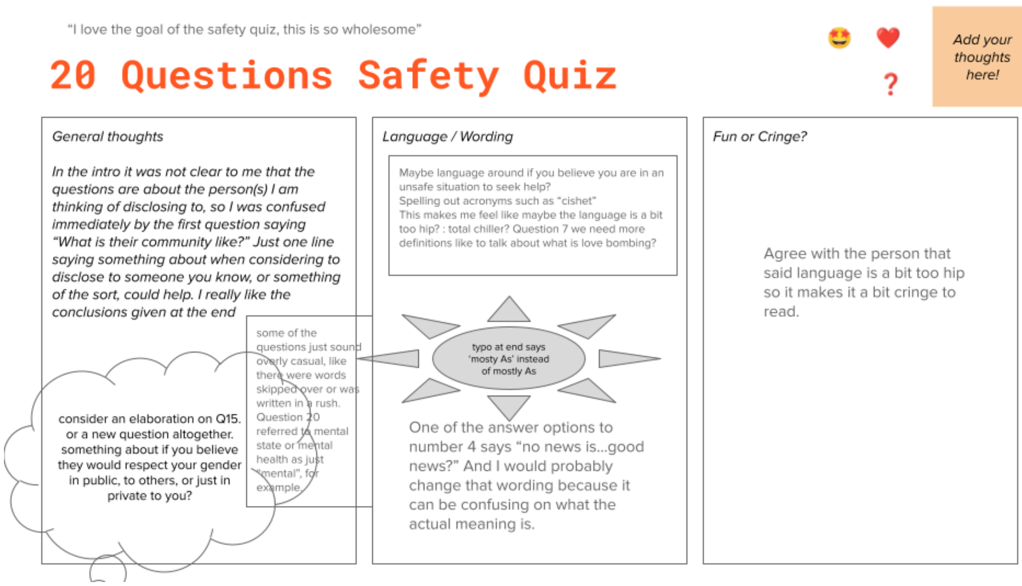


Fig. 3. Examples of how YAB members provided feedback during meetings as prototypes progressed further, leading to more specific feedback.

and added their own responses. Much like our co-design engagements with the YAB, at a high-level this process involved key phases of ideation generation, feedback elicitation, and collective reflection. Finally, we validated the design principles with five YAB members. After presenting the design values, the lead author asked YAB members whether and how they recognized each, leading to discussion of how to further improve the prototypes and refine the design principles.

3.3 Data Analysis

Our data consisted of YAB insights from participatory design activities in the form of written comments on study materials (e.g., sticky note additions to Google Slides), observational notes taken by research team members, and reflections generated in the retrospective value elicitation exercise. The research team analyzed each through a reflexive thematic analysis, synthesizing themes through an interpretivist, constructionist perspective [17, 96]. This position allowed the research team to identify and interpret patterns across the data while remaining aware of the our biases. We also drew from deductive frameworks to guide study activities, such as the six sexual health content areas that trans and queer youth seek out and four preferred sources of information.

The short windows between meetings necessitated rapid analyses in order to synthesize and create prototypes in time to present at the next gathering. Grouping themes in the data was primarily focused on identifying design needs and requirements. We sought to mitigate the limitations of this quickened pace through two approaches. First, eliciting feedback on each month's design and content development progress provided a regular opportunity for member checking. Common in community-based research, member checking involves sharing analyses with participants and validating researchers' interpretations [43]. In addition, the research, design, and content development teams frequently engaged in critical reflections of how our assumptions, values, and biases informed what we produced. For example, this directly informed our decision to engage in a reflexive value elicitation exercise. We acknowledged the extent to which our own identities as researchers, designers, and content developers overlapped and differed from those of the YAB. As our team is collectively made up of women, people of color, trans, queer, and non-binary people, we drew from our own experiences to make the content feel useful and realistic to the complex nuances of real-life situations. However, we remained aware that we were unable to fully comprehend how the YAB experienced the world, thereby emphasizing the importance of member-checking practices. Furthermore, commitments to impact-oriented, social justice-focused, and community partnership-driven research have informed the author team's efforts to design tools for trans and queer youth from the beginning.

3.4 Limitations

While our positionality offered both strengths and limitations, this study is further restricted in missing a final evaluation study. We had initially intended to conduct an evaluation study of the generated design principles and values from the retrospective value elicitation exercise, in which we would ask YAB members and additional trans and queer youth to validate whether they could recognize values in our prototypes. However, we were unable to complete this phase due to the early termination of our grant funding. In addition, while there were 18 active members part of the YAB, we faced difficulty with consistent engagement. Not every YAB member was active throughout the entire multi-year collaboration, as many withdrew or significantly lessened their participation over time.

Another limitation of this work stemmed from the importance of protecting YAB members' anonymity. Given the anti-trans political climate surrounding this work, it was vital to protect YAB members' personally identifying data from possible injunctions or data requests. We therefore made deliberate choices to significantly limit storing data that could be linked to their identities. We did not record any meetings and we documented all YAB feedback anonymously. For instance, YAB members wrote in Google Slides text boxes with their feedback, rather than leaving comments that were tied to their accounts. The data presented in this work are therefore not traced to individual YAB members and primarily draws from the researchers' perspectives, despite the participatory ethos sustained throughout the study.

Finally, as a single case study of a collaborative effort to design sexual communication tools for and with trans and queer youth, our work faces several limitations regarding generalizability and evaluative depth. Regarding the former, the partner YAB group is geographically constrained to the United States and not fully representative of all trans and queer youth. Therefore, their opinions likely do not reflect broader cultural contexts. Additionally, our understanding of the efficacy and satisfaction of our sexual communication tools is constrained to the YAB's perspectives. We had initially sought out a long-term evaluation but were unable to do so following the cancellation of federal funding. We therefore make no claims about the generalizability and broader impacts of our findings, but rather present potentially transferable insights as a case study that can provide guidance and inspiration for future work.

4 Findings: Overview of the Design Process

The YAB partnership yielded important insights into how to develop technology-based sexual communication interventions to support trans and queer youth with their sexual health. In particular, the needs assessment activities identified several design requirements for the tools we sought to build. First, technology-based interventions for sexual health should be interactive; given, as one YAB member stated, an "*app's memorability directly relates to its interactive features.*" They should also organize information in a clear way and feature a visual design that would appeal to trans and queer youth. One YAB member commented that an initial prototype was evoking a "*'sad beige' aesthetic that doesn't feel accessible or relevant to the population.*" YAB members also advocated for tools that users could engage with anonymously, given the importance of protecting data security and user safety for trans and queer youth. Finally, some individuals of the group were conscious of how potential bad actors might react and urged the group to make decisions based on how others might perceive the tools. As they shared, "*I find the use of the website to be the most exciting, but the inevitable weaponization of it to be less exciting. It's an unfortunate part of trans livelihood that most things we use or enjoy will end up in some [redacted] news article making it out as though we're indoctrinating children.*" Our initial exploration of how technology-based tools can support trans and queer youth with their sexual health thus emphasized interactivity, privacy and security, and safety, providing the foundation for the subsequent design-focused phase.

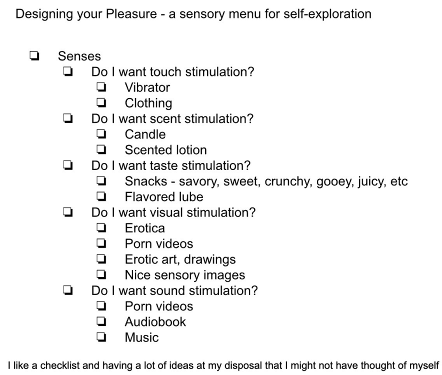
Our discussion of how to assist trans and queer youths' sexual health particularly highlighted the importance of supporting their ability to discuss important topics like STI information, desires, and boundaries with others. The design partnership with the YAB thus yielded two online tools that supported sexual communication between trans and queer youth and their partners. First, Desire Dialogues features a survey that scaffolds both individual reflection of sexual boundaries and collaborative discussion with their partner(s). Next, the Intimate Disclosure Toolkit presents a set of reflection- and simulation-based activities to prepare users to disclose personal information with their partners. In the following section, we outline how we identified each design need, developed prototypes to address them, then sought to understand to what extent we addressed the need through YAB evaluation. All presented quotes are anonymized and either paraphrased from meeting discussions or taken directly from group materials.

4.1 Desire Dialogues: A Three-Part Survey to Jumpstart Communication

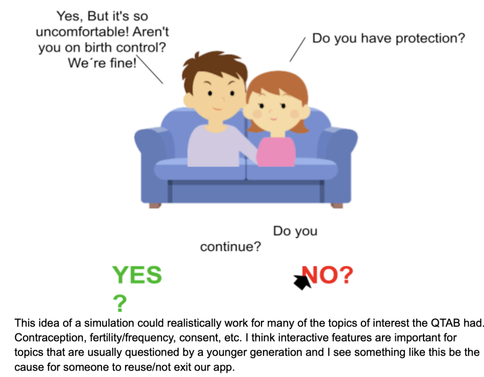
The first thematic area constructed with participants focused on the need to support trans and queer youth with communicating their sexual boundaries, interests, and preferences with their partner(s). Given the lack of focus on consent and pleasure in youth sexual communication literature [115], this YAB-initiated direction proved valuable.

YAB members commonly discussed the value of sexual communication around social and relationship dynamics. For example, YAB members named "*exploring healthy relationship dynamics*", "*engaging in polyamory w/out proper communication (or any relationship)*", and "*navigating intimate relationships, [especially] with those unfamiliar with trans bodies*" as current sexual health-related challenges trans and queer youth face. In the brainstorming exercise, one YAB member proposed the idea of a "*quiz of what you might be interested in [with an] option to compare with a partner's responses.*" YAB members built upon this idea by sketching how a survey might facilitate sexual communication. Figure 4a depicts a checklist that users can use to determine what they are interested in. Figure 4b simulates a sexual encounter where they are being pressured to not use contraceptives, prompting the user to decide whether they should continue or not. YAB-generated storyboards captured how and why a user might engage with the survey, as one member described the survey as "*informative but sexy foreplay*". In Figure 5, another YAB member outlined how the survey can help someone who may be new to dating "*to prepare themselves for communication*", "*use the survey to help determine what they feel comfortable doing sexually*", then prepare to discuss their newfound understanding with a partner. We thus identified that sexual communication tools should facilitate individual reflection, provide practice navigating potentially uncomfortable situations, and establish confidence in the user's preferences and their ability to communicate with their partner(s).

4.1.1 Overview of Desire Dialogues. The Desire Dialogues prototype features a thirty-nine question survey split into three parts: first, individual reflection over one's own boundaries; next, individual preparation for communicating these boundaries; and finally, preparation to facilitate discussion



(a) YAB sketch of a self-exploration sensory menu for understanding personal interests.



(b) YAB sketch of a simulated conversation with a partner involving pressure to not use contraceptives.

Fig. 4. YAB sketches to support communicating sexual boundaries.

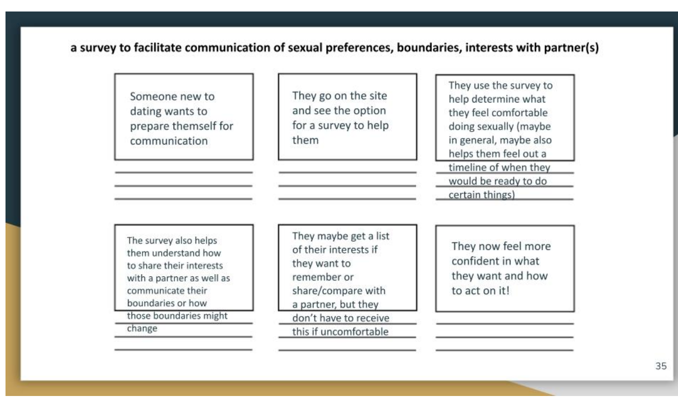
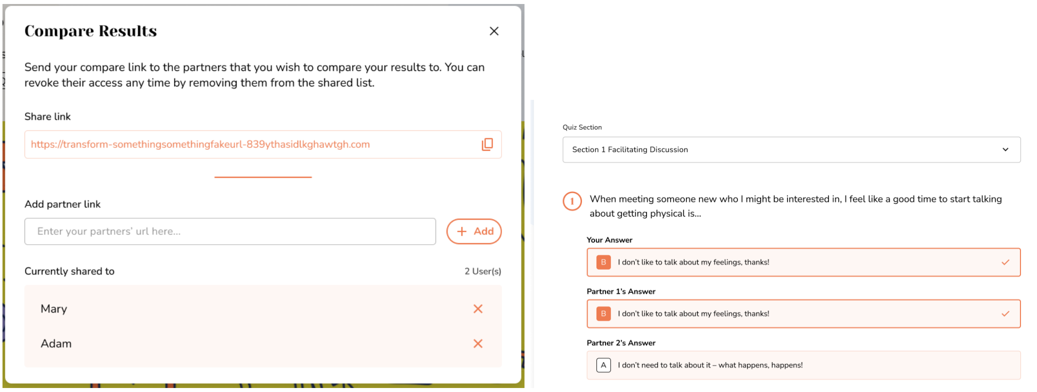


Fig. 5. YAB members created storyboards to depict how a user might engage with a survey for communicating their boundaries and interests with their partner(s).

with their partner(s). Appendix A outlines sample questions for each section. When the user has completed the survey, they can download their responses and receive a unique link. Survey takers can then compare their results with their partners' to view what questions they agreed upon and where they diverged, serving as a visual tool and prompt for collaborative discussion, seen in Figure 6.

4.1.2 Desire Dialogues Evaluation. YAB feedback was largely in favor of the prototype, sharing that the survey aligned with their vision of the tool established from the needs assessment discussion. One YAB member described the value of this tool as "*very useful to facilitate communication about things that you would otherwise be nervous to talk about or maybe you wouldn't have the words or confidence to bring up [. . .] this is like the frame for that conversation [...] it can be difficult to have a conversation like this without that*". Another appreciated the option to include multiple partners in the comparison feature, writing "*love the inclusion of poly couples and its super cool that you can have more than two people*".



(a) After completing the survey, users receive a link for comparison results with their partner(s).

(b) The resulting comparison outlines how each person's responses align or differ for each question.

Fig. 6. Overview of Desire Dialogue's comparison feature.

YAB members also provided constructive feedback for how to improve the prototype. For example, several agreed that the survey should provide a disclaimer that the results are not a precise indicator of whether people are a good match and that "*a low score doesn't mean you aren't compatible*". Most critically, YAB members had concerns around the comparison feature's potential for breaches of privacy and security. As one member shared, "*sometimes I send links without realizing I've pasted the wrong thing [...] so thinking about if someone sends the link to the wrong person [. . .] if it's a unique code, then it would be more secure*". As a result, we iterated upon the comparison feature by requiring passwords for sharing and including text that emphasized the importance of trusting who they shared their responses with.

This prototype therefore responded to several of the YAB design needs. The survey is designed to be interactive via its teen magazine quiz-like style and users can complete it privately without providing any personal information. Questions were also intended to guide users through a reflection process, prompting them to consider their own communication patterns and how they feel about various experiences (e.g., what happens when their boundaries are crossed or their comfort level with sexting). Saving, sharing, and comparing results also provides a starting point to discuss boundaries with their partner(s). Ultimately, what began as an idea from a YAB member to build a "*quiz of what you might be interested in [with an] option to compare with a partner's responses*" was created into an interactive prototype known as Desire Dialogues.

4.2 Intimate Disclosure Suite

The second thematic area constructed was around the need to support trans and queer youth with disclosing their identities. Kosenko [59] describe how disclosure presents a communication dilemma for trans people, as disclosing one's identity before a sexual encounter can reduce the risk of physical harm, but it can also lead to rejection. YAB members affirmed this dilemma, discussing the tensions they faced when disclosing personal information with others. When reflecting on their experience with disclosure, one YAB member shared: "*disclosure depends on the environment [and] reaction[s] from strangers [are] usually indifferent but family members tend to react poorly*". As another YAB member advised, "*have support ready. Despite people knowing you for so long, there can be at times a sense of grief or loss of the person you once were in their eyes.*"

Fig. 7. YAB-generated sketch of how to support disclosure.

Addressing safety when disclosing was therefore a primary concern for the YAB. In response, YAB members brainstormed surveys that assess the safety of a potential disclosure scenario, describing "q*uiz-like feature that can give a person scenario outcomes and how to deal with them, as well as prior questions to ask to ensure safety in coming out."* YAB members also sketched a tool that helps users deal with hateful reactions when coming out, suggesting that it "*give constructive ways to get out of those situations/ease them*", seen in Figure 7. Most sketches related to supporting disclosure featured simulations of conversations and thinking ahead about what one might do in different scenarios. Therefore, navigating the disclosure process safely and capably surfaced as a primary need that a technology-based intervention could support.

Furthermore, disclosure involves more than the act of sharing information with others; it is a deeply personal experience that requires a person to reflect and understand what they want to disclose and why. Many group members advised that others "*consider what your goals are with sharing your identity*" and "*consider what your reasons are for doing so before proceeding*". Others indicated that internal pressures can arise in the process, such as "*potential stigmas or feelings of guilt*". The YAB therefore revealed that a technology-based intervention should also account for the emotional individual experience of disclosure.

4.2.1 Overview of Intimate Disclosure Toolkit. We initially sought to develop a toolkit to support several forms of disclosures including those with family members and healthcare providers; however, we chose to focus on disclosure with intimate partners due to resource and time constraints. The Intimate Disclosure Toolkit consists of six activities, seen in Figure 8. In Prepare, users answer questions to gauge how prepared they may feel disclosing their gender identity. In Reflect, users plan ahead by reflecting on what information they wish to disclose. In Assess, users evaluate the safety conditions of a potential gender identity disclosure to a specific person. In Discuss, users practice conversations with others through simulated real-world scenarios (e.g., flirting with a dating app match or updating a casual relationship about new pronouns). In Respond, users practice responding to potential conversations through various decision-based scenarios. Finally in Interpret, users translate common responses and scenarios to help them learn about their own boundaries. Four activities feature quiz structures (See Figure 9b as an example). Discuss features a

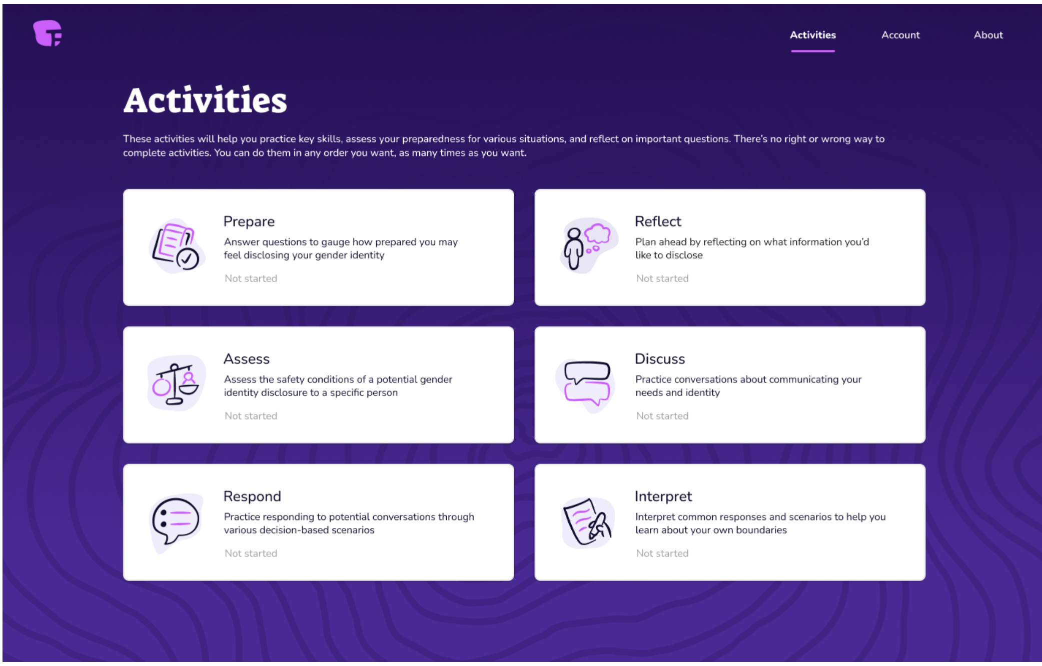
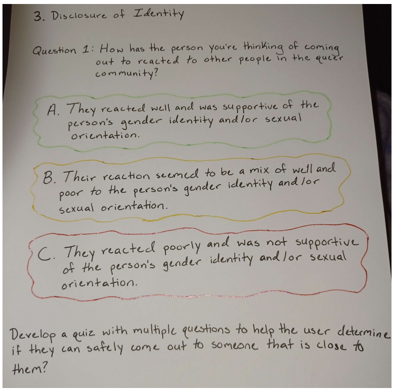


Fig. 8. Overview of the six activities part of the Intimate Disclosure Toolkit.

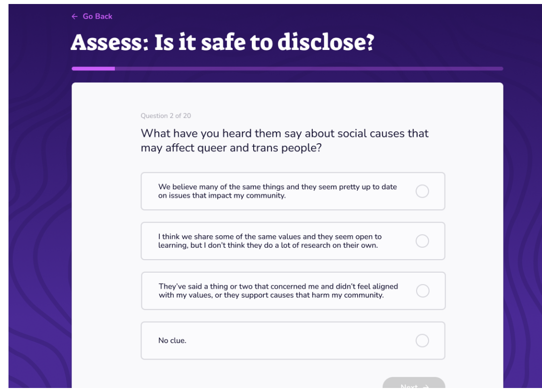
dating simulator-style interaction where users can select how they wish to respond, and Interpret asks users to classify scenarios into red and green flags. Please see Appendix B for examples the activities.

4.2.2 Intimate Disclosure Toolkit Evaluation. YAB members were enthusiastic about the Intimate Toolkit activities while offering points of clarification. As one member described: "*I really like the self assessment piece, especially those with mixed or complex emotions about discourse could help start unpacking that*". Another appreciated the tool's capacity to prepare users for how people might respond, especially in anticipation of harmful reactions. Others crucially cautioned about being overly prescriptive about the process of intimate disclosure: "*We need to keep in mind not making queer identities, relationships, break ups, etc. into monoliths. Everyone's identity, sexuality, culture, family history, etc. comes into play and we need to be sure not to make one experience the 'norm' and make others on the fringes*". As another noted, "*different people have different flags.*"

The research, design, and content development teams specifically sought feedback on whether the toolkit's language came off as accessible to trans and queer young people. We originally introduced a casual tone that would appeal to young people, but YAB members held different opinions about the efficacy of doing so. One person shared "*I think most of it is fun, I think some of the language or situations could be a little looser. In some places it still feels a little PSA-ish*", while another disliked the youth-appealing language by saying they "*agree with the person that said language is a bit too hip so it makes it a bit cringe to read.*" To provide guidance for striking the right tone, one person shared that "*friendliness is more important. People are coming to this as a resource, as a source of information. Everyone expects a certain level of professionalism. When we go to educational resources, we have some level of expectation... even kids want adults to talk to them like they are a person.*"



(a) YAB-generated sketch of a survey to assess the safety of a potential disclosure scenario.



(b) A sample question from the Intimate Disclosure Toolkit's Assess activity.

Fig. 9. YAB ideas laid the groundwork for the activities and features in the Intimate Disclosure Toolkit.

Thus, future tools should thus employ a friendly tone and language, rather than guessing how a young person might talk.

The Intimate Disclosure Toolkit ultimately targeted needs surfaced from early needs assessment sessions. The quiz-based activities guide individual reflection by scaffolding consideration around what users want to disclose, for what reasons, and how they want the interaction to go. In doing so, we intended to emphasize that users have agency over how their disclosure experiences happen, contrasting the lack of control over social situations that trans and queer youth commonly face. Users can additionally establish preparedness and confidence in navigating future scenarios through activities that simulate real-world conversations in Discuss, provide options for how they can respond to different reactions in Respond, and consider safety plans (e.g., how they might get home safely if a disclosure does not go well) in Assess. One YAB member shared their appreciation for how the toolkit "*showed examples of setting boundaries, and the differences between what assertive/aggressive communication looks like.*" In the end, YAB ideas generated the activities that comprise the Intimate Disclosure Toolkit, as shown in Figure 9.

5 Design Principles

Following the design process, the content and design teams engaged in a retrospective value elicitation exercise to identify motivations that drove specific decisions, which were then validated by YAB members. From this discussion, we outline five design principles that guided our collaborative process: approaching safety through a non-prescriptive, informed consent model; translating a joyful user experience into joyful signals; reinforcing individual strengths and readiness; support for real-world possibilities; and suspending designers' assumptions.

5.1 Principle 1: Approaching safety through a non-prescriptive, informed consent model

Safety is a crucial part of sexual health, and we made several decisions to protect users. First, none of the features took in or stored personally identifiable information to protect against potential security violations. Next, we emphasized user agency in their decision making and avoided making recommendations. This motivated creating features that supported users to assess risk for themselves by identifying potential red flags, safe spaces, trusted support networks, and their own ways they would handle potential conflicts. We additionally sought to present a diversity of outcomes with a level of neutrality and nuance, because we recognized that most sexual communication situations are not straightforward and we wanted youth to make their own determinations. This approach is especially important when dealing with sensitive issues like disclosure.

This principle echoed YAB-provided suggestions. As one YAB member advised, "*[the tool] should guide the person to possible choices they might make–not directing them how to act but rather providing guidance based on their personal situation. It should allow someone to make their own decisions but feel less overwhelmed with the daunting thing they're trying to do.*" In practice, this non-prescriptive principle manifested itself in places such as the decision tree part of the Intimate Disclosure Suite. Here, the tool offers multiple ways a person can react and respond to a scenario. For instance, they can choose to respond with humor, education, or confrontation; pause & reflect by taking time for themselves, consulting a friend, or ignoring it altogether; and end the interaction by leaving or setting a boundary. One YAB member found this aspect useful, stating how "*there's several layers of multiple choice which gives the user ability to explore options of confrontation and considerations of safety*". In the same toolkit, the safety assessment quiz provides 20 questions a person can answer. At the end, the quiz provides reflection prompts for "mostly a's—green flags, mostly b's—yellow flags, mostly c's—red flags, mostly d's—need more information." While those red, yellow, green flags are our determination, we avoid giving an exact number and let them decide how to determine "mostly" for themselves. One YAB member appreciated "*how the narration showed that in one of the scenarios there was no "wrong" answer but instead explaining that both choices could work*".

5.2 Principle 2: Centering joyful and interactive experiences

While we recognized the importance of teaching about the dangers of romantic and sexual relationships, it was also vital to balance this message with pleasure and joy. We sought to capture scenarios that were fun and not only reflective of challenges or issues people may encounter while dating or exploring. By centering a joyful, engaging user experience, the tools would be more valuable for users, which led us to explore different interaction techniques like a dating simulator-style activity for modeling discussion, a red/green flag scroll scale for interpreting responses, and the ability to connect with others by comparing with other results in Desire Dialogues. YAB members agreed that the interactive tools were exciting to use. As one person described them, they provided "*enjoyable, low stakes way of having that conversation with somebody*".

5.3 Principle 3: Reinforcing individual strengths and readiness

The initial motivation behind the development of these interactive features was around preparation for potentially complicated, future sexual communication experiences. As part of this, it was important to be honest about potential reactions and preparing users for all types of consequences. We also sought to help users forge the skills necessary to handle whatever might happen. Many of the surveys were designed to remind the user that they are the one making the choices and they can determine what is best for them. We simply provide the prompts of things they may want to consider while they identify their own needs. Several YAB members pointed to the self-reflection activity of the Intimate Disclosure Toolkit as an example of emphasizing individuals' control over what they disclosed or not. The survey is designed to adapt to however much a user wants to share and never pushes the user to disclose information they might not be comfortable with. One YAB member agreed that the Reflect activity captured this principle well, sharing "*I think that it does a good job at capturing several different aspects of relationships and different scenarios where you might not even know what to say. But if you did this activity, it would help you formulate what you would want to say if you were in that situation, so the readiness aspect is there.*"

5.4 Principle 4: Support for real-world possibilities

We recognized how typical, over-generalized sex education often ignores the specific experiences that trans and queer people go through. Therefore, we sought to incorporate situations and dialogue that users could identify with to ensure that products felt reflective of peoples' actual lives. YAB members valued "*real*" and no fluff in the processes, preferring conversational activities to help them prepare for realistic scenarios. This took form in the Modeling Discussion component of the Intimate Disclosure Suite, which gives three scenarios that encapsulate different kinds of potentially intimate interactions: chatting with a flirty barista, responding to a friend's advances, talking to a situationship. A YAB member appreciated the realistic nature of this activity, adding that it "*goes into scenarios that are everyday and might happen to you randomly [...] if I was taking that activity I would feel more prepared for a real world interaction*". Multiple YAB members also indicated that Desire Dialogues' capacity for multiple partners as a positive example of this principle.

5.5 Principle 5: Suspending designers' assumptions

Finally, team members emphasized the importance of being aware of their own self-identities in the process and how they may influence the decisions that we may make in designs. The teams also advocated for an awareness of what they did not know or could not speak to. We commonly discussed what we did not know, especially as it pertained to trans and queer youths' experiences today. As a result, each interactive feature was designed to provide youth with the tools to know themselves better, rather than directing them on what to do based on a false presumption that we as the development teams are all-knowing. YAB members felt that this principle manifested throughout both tools. Many shared that the sheer depth of the activities indicated that the tools provide space for users to fill in gaps, rather making assumptions of their experiences.

6 Discussion

Designing the two technology-based sexual communication tools, Desire Dialogues and the Intimate Disclosure Suite, required a reflexive, collaborative, and iterative YAB partnership and the five design values embedded in decision-making processes. As a result, we advance scholarship on supporting sexual communication between youth and their partners by pursuing technology-based interventions, facilitating collaboration around relational topics in sexual boundaries and disclosure [115, 116]. In the following section, we engage with the opportunities and challenges of design partnerships, thereby clarifying ways to co-create future technologies that promote sexual communication among trans and queer youth.

6.1 Reflection of the Method

From the onset, the research team sought to establish true collaborative partnerships with trans and queer youth. We made this clear when recruiting YAB members and consistently emphasized our commitments throughout the multi-year project. However, as many scholars have documented, there are often gaps in participatory work between what researchers intend to accomplish and what is feasibly achieved [7, 92]. Like previous participatory work, we encountered numerous challenges and success in achieving what Baldauf et al. [7] describe as "genuine participation", culminating in lessons for future work.

The primary challenge we faced was in achieving "genuine participation". Beginning with the difficulties of establishing true participation, our original goal was to align YAB members' positions as design partners, but their involvement drew closer to the informant role described by Druin [32]. YAB members consulted on the project, then design, content, and research teams synthesized and executed ideas [46]. Despite yielding valuable insights, it is possible that our participation arrangement did not fully enable YAB members to voice their concerns. This gap was in part due to the at times competing priorities among each team involved in the project. Professional expectations such as contracts and academic timelines as well as structural constraints like funding structures motivated the research, content development, and design teams to continuously push work forward. At times, this led to driving the YAB to make decisions rather than spend more time deliberating on the best direction to take. These forces also shaped the sexual communication tools. For example, we were forced to constrain the disclosure toolkit to a specific context, intimate partners, even though the YAB explained that disclosure was a much more expansive experience. Due to resource and labor constraints, we were unable to generate the full tool that the YAB sought out.

Another contributing factor was that the partnerships we sought to establish added labor and time burdens onto YAB members. As previous scholars have indicated [46], youth must balance study tasks in addition to their everyday responsibilities, like navigating school and work. One YAB member called into a meeting during their school's lunch period and others often requested asynchronous participation options due to their overwhelming schedules and lives. In dealing with these challenges, we remained deeply engaged with the YAB members to produce the sexual communication tools and were transparent about why we made decisions. For example, when constraining the disclosure toolkit, we explained that our remaining funds could not support the contract required for the content development team to build the fuller version. We thus approached each challenge rooted in the same participatory ethos that has supported this work from the beginning, demonstrating that future researchers may never be able to achieve true partnerships, but they can navigate each challenge collaboratively and honestly.

While equal participation was difficult to fully achieve, our collaboration and the resulting sexual communication tools benefited from desire-based design frameworks and theories of transformative reflection. First, *desire-based design* manifested in how we engaged with the YAB and the specific decisions behind the generated tools. Existing frameworks distinguish between what people need and what they want [75, 101]. Nelson and Stolterman [75] capture why this distinction is valuable, writing that desire acts as a "destabilizing trigger for transformational change, which facilitates the emergence of new possibilities and realizations of human 'being'". Desire motivated exploration into how to create engaging features that trans and queer youth would feel excited to learn from, leading to the different interaction styles present across the tools (e.g., survey, slider scales for determining red/yellow flags, swiping). In our interactions with the YAB, we additionally centered what they desired in how they connected with others, allowing researchers to understand participatory partners more fully. We learned how YAB members did not only want to be free of STIs and harm; they wanted to experience close and meaningful connections with others too. Discussions grounded in desire additionally sparked YAB members to open up about how their trans and queer identities informed their experiences with relationships and how they communicate with their partners. Várhidi and Rauhut [104] took a similar design approach when creating a digital toy for communicating people's desires that paired perspectives on infection and reproduction with playfulness, pleasure, self-discovery, and communication. Drawing focus to desire, rather than simply what people need to survive, can thus open to newer, deeper discoveries that benefit the design process.

Future social computing research can therefore benefit from incorporating desire into their processes. This particular design orientation can benefit the uptake and adoption of the digital tools researchers develop. Engaging, desire-driven content that combines interactive elements with accurate sexual health info has been an effective strategy for meeting young people's sexual health information needs [62, 70, 119]. However, in recommending an orientation towards desire, we also caution against ignoring the unfair realities that many marginalized people face. Trans people have much to be angry about, and trans rage has motivated active resistance against systems of oppression [13, 48, 66]. Rage and desire, then, can work in tandem to support the development of sexual communication systems, and it is incumbent upon researchers and technologists to deeply understand the experiences of the people with design for and with.

In addition, *transformative reflection* was at the center of the tools the YAB desired. Both tools provide strategies for individuals to reflect upon their needs and desires, then decide how to proceed. Reflection in this way follows Design Principle 1 to provide a self-guided, non-prescriptive process that targets support individually, by facilitating a deeper understanding of the self, and more broadly, by fostering preparedness for navigating the social realities of being a trans or queer young person. While this form of reflection is centered around the individual, it provides necessary work before people engage with others, indicating that sexual communication technologies can benefit from scaffolding transformative reflection.

Our strategies for supporting transformative reflection draw from past HCI research. In particular, Schön's concept of the reflection practicum identifies strategies to scaffold technology-mediated reflection for transformative change [97]. A reflection practicum structures learning by allowing people to "experiment at low risks, vary the pace and focus of work, and go back to do things over when it seems useful to do so" [95]. Fleck and Fitzpatrick [37] have also identified techniques for supporting multiple levels of reflection through recording knowledge and experience, prompting explanations, and interpreting situations through multiple perspectives. We enacted these transformative reflection lessons in various ways. Both interactive features support recording knowledge, prompting explanation, and seeing situations through different lenses. For instance, Desire Dialogues provides a way for partners to discuss their overlapping and/or mismatched perspectives on their boundaries and interests, Respond in the Intimate Disclosure Suite allows users to explore various reactions (respond with humor, aggression, or education; pause and reflect; and end the interaction), and all features provide savable summaries of their responses. We further provide safe testing grounds to test different ways of responding to situations. In the Discuss activity of the Intimate Disclosure Suite, users can choose different responses posed by an imagined audience. The tool does not prescribe correct or incorrect answers but instead provides coaching for thinking about how they might consider the interaction. Given scholars have explored the role of reflection in health contexts, [14, 67, 85, 86], **transformative reflection offers tangible strategies for meaningfully translating individual considerations into collaborative action**.

6.2 Future Directions of Sexual Health Technologies for Social Computing

While this paper focuses on sexual communication between trans and queer youth and their partners, there are transferable lessons into designing future technology-based interventions for sexual health more broadly. Social computing research is well-positioned to make meaningful contributions to this space. As we have shown through this work, technology-based interventions benefit from their capacity for privacy, personalization, and interactivity. Further, sexual health is of individual and collective importance, aligning efforts to promote sexual health with collaborative investigations core to Social Computing. Sociologist Steven Epstein elucidates that sexual health acts as a site of various overlapping sociopolitical disputes and a pathway into enacting social justice for all. As he argues, attending to sexual health is a significant and worthwhile pursuit because "sexual health projects have expanded popular understandings of what may be encompassed under notions of human rights and social justice. Such efforts do more than cure disease, as important as that may be: '*they expand the scope of freedom and capacity for fulfillment available to many individuals*' [35, p 9, emphasis added]. Sociologist Héctor Carillo further offers that examining sexuality provides a perspective to "shed light on culture, power, social interaction, social inequality, globalization, social movements, science, health, morality, and public policy" [21]. As these critical

Level	Opportunities	Summary
Individual	Reflection Preparedness Self-Efficacy	Sexual health technologies can enable individual reflection, preparedness, and self-efficacy. Here, we define self-efficacy as a person's perceived capacity to navigate their own challenges [8]. In our work, for example, Desire Dialogues facilitates reflection so that individuals can better understand themselves, their needs, interests, boundaries, and desires. Features part of The Intimate Disclosure Suite like Assess guides preparedness for potential outcomes by emphasizing emotional and physical well-being and establishes self-efficacy in knowing themselves and their readiness for potential interactions (e.g., Discuss, Interpret, and Respond Activities part of the Intimate Disclosure Toolkit).
Community	Communication Connection to Community	Sexual health technologies can provide mechanisms for communication and connection with others. For instance, the Desire Dialogues feature provides a way to compare and discuss survey results with a person's partner(s). Other scholars have developed conversation agents to facilitate relationship communication to promote "safe" sex [6]. An additional body of work has examined how online spaces provide space for community connection and support (e.g., Chuanromanee et al. [24] describes how transmaculine and non-binary users sought out decision-making support for pursuing top-surgery).
Structural	Education Destigmatization Access	Sexual health technologies can address structural forces that facilitate or hinder health and well-being. Technologies that provide information can educate people through comprehensive, inclusive, and affirming sex education [62]. They can also destigmatize taboo topics, as scholars have shown Tiktok memes have been able to accomplish when it comes to sexual health myths [110]. Sexual health technologies can also promote access. Researchers have identified design strategies that extend access to sexual health information for blind and partially sighted people as well as how sexual health technologies can provide care when there are barriers to accessing in-person care [72, 81].

Table 1. Leveling Up Opportunities for Sexual Health Technologies in Social Computing Research

sexuality scholars have raised, the realm of sexual health's societal impact is vast; therefore, it is vital to draw attention to intentional, value-based, and community-driven technology development that advances sexual health for all. Despite its social nature, however, sexual health research is notably scarce in the Social Computing literature.

Given its importance in advancing overarching facets of society, we draw from our own project as well as those found in the literature to outline future work possibilities for sexual health technologies, seen in Table 1. Drawing inspiration from Wesp et al. [111]'s conceptual work for Intersectionality Research for Transgender Health Justice and Veinot et al. [105]'s mapping of upstream health informatics interventions for health equity, we map opportunities onto individual, community, and structural levels as potential areas for intervention. In this work, we demonstrate how sexual communication technologies can work to mediate communication and support decisions, two areas that Veinot et al. [105] identify as potential approaches to reduce health inequities. Future work might take aim at additional areas outlined in Table 1 while integrating a similar participatory approach outlined in this paper.

These lessons are especially significant for future contexts that use Artificial Intelligence (AI) tools to address sexual health topics. Current trends point to growing use of generative AI and Large Language Models for sexual health education and counseling (e.g., [23, 31, 57, 74, 79]). Without clear guidance of how to work with communities to build technology-based interventions that meet people's needs or why these innovations are valuable, investing in these AI-based systems can end up worsening the very issues they seek to address, known as intervention-generated inequities [106]. Therefore, this work seeks to ground future development in participatory partnerships to reach Epstein and Carillo's visions for both advancing sexual health and transforming human rights and social justice.

7 Conclusion

By outlining our collaborative design process of two technology-based sexual communication tools in Desire Dialogues and the Intimate Disclosure Toolkit, future development can draw from our partnership model, reflections on the challenges of engagement, and integration of desire-based frameworks and transformative reflection. Additional strategies for designing future technology-based sexual communication tools can build from the five design principles identified in this work—approaching safety through a non-prescriptive, informed consent model; translating a joyful user experience into joyful signals; reinforcing individual strengths and readiness; support for real-world possibilities; and suspending designers' assumptions. In summary, our case study documenting the participatory challenges and successes encountered in this work contributes further understanding into co-creating technology-based sexual communication tools for and with trans and queer youth.

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A Desire Dialogues Sample Questions

Phase 1: Individual Reflection

When meeting someone new who I might be interested in, I feel like a good time to start talking about getting physical is…

(1) I don't need to talk about it – what happens, happens!
(2) As soon as mutual attraction has been established.
(3) After a few weeks of going on dates and/or messaging back and forth.
(4) I need to feel like I know someone really well before we even go there.
(5) I don't know! It really depends on the person or people involved.

What expectations do I have of people I am having sex (or physical intimacy) with? (Check all that apply)

(1) None!
(2) Good and consistent communication.
(3) Regular check-ins about STI testing and other partners.
(4) No physical intimacy until there is an emotional connection.
(5) No physical intimacy until there is a relationship commitment.

(6) I don't know yet!

Exchanging sexy messages through text or online (sexting)

(1) I want to try this.
(2) I want to avoid this.
(3) I want to learn more about this.
(4) This is okay all the time.
(5) This might be okay, but we have to talk about it beforehand each time.

Phase 2: Individual Preparation

What is my ideal way to discuss sexual boundaries?

(1) Schedule a time to have a serious face-to-face about hard lines, curiosities, and desires.
(2) A text conversation.
(3) Exchanging fantasies to focus on what we each *do* want – keeping it more sexy than serious.
(4) I'd prefer to just let things come up naturally.
(5) I'm not sure.

What will I do if my partner doesn't react well to my boundaries?

(1) Explain to my partner that I'm not comfortable getting sexual.
(2) Schedule a time to have further discussion.
(3) Ghost the person to avoid uncomfortable conversation, maybe even danger.
(4) I know I have people pleasing tendencies and I may struggle to advocate for myself.
(5) I'm not sure.

If I'm feeling anxious, what will be my go-to?

(1) Slow down what's happening and focus on breathing.
(2) Figure out if I've felt this way before in other situations and if so, use strategies that have worked for me in the past.
(3) Acknowledge the feeling for myself but not necessarily take action.
(4) Tell my partner about it and then decide how to proceed.
(5) I'm not sure.

Phase 3: Facilitating Discussion with Partner(s)

What is your main goal for this conversation?

(1) Aligning our expectations.
(2) Breaking the ice regarding topics sometimes difficult to discuss.
(3) Getting to know ourselves and each other better.
(4) General practice in communicating about sex and boundaries.
(5) I'm not sure.

What is your preferred communication method for having difficult conversations?

(1) Talk immediately as issues arise.
(2) Take space and schedule a time for face-to-face discussion.
(3) Take space and schedule a phone call.
(4) Text, email, or letter.
(5) I'm not sure.

B Intimate Disclosure Toolkit Sample Questions

Readiness Assessment: Are You Thinking About Disclosing?

Why are you considering disclosing?

(1) For my own benefit (authenticity)

(2) For the other person's benefit (feel like I'm 'lying', etc.) (afraid of what might happen if I don't)

When are you thinking about disclosing?

(1) Before we make plans
(2) During the plans
(3) With a call or text after the plans if they go well

The Self-Reflection Section: What do you want to disclose?

What's okay to talk about with others – think about what you're okay with being shared outside the relationship and whether that should be part of your disclosure process. For example, you may feel fine with a potential partner disclosing your gender identity to people they know (e.g., "Alex is trans") but not medical information ("Alex is getting bottom surgery"). It also may be okay to share information with some groups (like friends) but not others (like coworkers).

(1) HELL YEAH – this feels important to share.
  (a) If selected, when will you share this?
    (i) I'll share this when I first start talking to someone.
    (ii) I'll share this within our first couple times hanging out.
    (iii) Not sure when I'll share this, but it will be before we get sexual and/or feelings are involved.
    (iv) I might hold onto this info for a while but share eventually if things get serious.
  (b) If selected, how much do you want to share?
    (i) I'll test the waters and decide based on how they respond to other information.
    (ii) I'll share the basics but hold back some of the details for now.
    (iii) I'm an open book! I'll share as much as they want to know.
    (iv) It's really important to me that they fully understand this so I'm going to make it a point to share as much as I possibly can.
(2) HMMM – I'll have to think about this more.
(3) NAH – I don't think I'm ready to go there (or this doesn't apply to me).

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